

C3-2

1. (a) 证明 $\hat{Q} = \sum_n q_n |e_n\rangle \langle e_n|$

设 $|\alpha\rangle = \sum_n c_n |e_n\rangle$, 则 $c_n = \langle e_n | \alpha \rangle$

又有 $\hat{Q} |e_n\rangle = q_n |e_n\rangle$

$$\begin{aligned}\hat{Q} |\alpha\rangle &= \hat{Q} \sum_n c_n |e_n\rangle \\ &= \sum_n c_n \hat{Q} |e_n\rangle = \sum_n \langle e_n | \alpha \rangle q_n |e_n\rangle \\ &= \sum_n q_n |e_n\rangle \langle e_n | \alpha \rangle\end{aligned}$$

则 $\hat{Q} = \sum_n q_n |e_n\rangle \langle e_n|$, 得证.

(b) $f(\hat{Q}) = \sum_n f(q_n) |e_n\rangle \langle e_n|$

则 $e^{\hat{Q}} = \sum_n e^{q_n} |e_n\rangle \langle e_n|$

$$\begin{aligned}&= \sum_n \left(1 + q_n + \frac{1}{2!} q_n^2 + \frac{1}{3!} q_n^3 + \dots + \frac{1}{n!} q_n^n \right) |e_n\rangle \langle e_n| \\ &= \sum_n |e_n\rangle \langle e_n| + \sum_n q_n |e_n\rangle \langle e_n| + \sum_n \frac{q_n^2}{2!} |e_n\rangle \langle e_n| \\ &\quad + \dots + \sum_n \frac{q_n^n}{n!} |e_n\rangle \langle e_n| \\ &= 1 + \hat{Q} + \hat{Q}^2 + \hat{Q}^3 + \dots + \hat{Q}^n \\ &\quad \uparrow \text{单位矩阵}\end{aligned}$$

2. $|b_n\rangle = \hat{U} |a_n\rangle$, 证明 $\hat{Q}_{(b)} = \hat{U} \hat{Q}_{(a)} \hat{U}^\dagger$

$\hat{Q}$ 的平均值: $\langle b_n | \hat{Q}_{(b)} | b_n \rangle = \langle a_n | \hat{U}^\dagger \hat{Q}_{(b)} \hat{U} | a_n \rangle$

$$\langle a_n | \hat{Q}_{(a)} | a_n \rangle = \langle b_n | \hat{Q}_{(b)} | b_n \rangle = q_n$$

$$\text{则 } \hat{U}^\dagger \hat{Q}_{(b)} \hat{U} = \hat{Q}_{(a)}$$

所以 $\hat{Q}_{(b)} = \hat{U} \hat{Q}_{(a)} \hat{U}^\dagger$, 得证